The HHL Algorithm

Assessment of the Harrow–Hassidim–Lloyd algorithm for $A\vec{x} = \vec{b}$

QUESTION PAPER

Time allowed:	3 hours	**Maximum marks:**	240
Questions:	80 (4 sections × 20)	**Answer key:**	separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Linear Algebra and Mathematical Foundations, **B** Quantum Computing Foundations, **C** HHL Algorithm Fundamentals, **D** Advanced HHL and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. Mark your answers on the **Response Sheet** at the end of this paper by darkening ONE bubble per question.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- Linear system $A\vec{x} = \vec{b}$, A a $d \times d$ matrix, $d = 2^{n_b}$. A Hermitian with eigenpairs $A\,|u_j\rangle = \lambda_j\,|u_j\rangle$; $|b\rangle = \vec{b}/\|\vec{b}\| = \sum_j \beta_j\,|u_j\rangle$; $|x\rangle = A^{-1}\,|b\rangle\,/\|A^{-1}\,|b\rangle\|$. Condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (Hermitian A), in general $\kappa = \sigma_{\max}/\sigma_{\min}$. Sparsity s = largest number of non-zero entries in a row.
- Basis states are written with **qubit 0 first = most significant bit**: $|q_0 q_1 \ldots\rangle$, e.g. $|10\rangle = |2\rangle$. $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_Y(\theta) = \left(\begin{smallmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{smallmatrix}\right)$; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. $\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{N}}\sum_y e^{2\pi i\,xy/N}\,|y\rangle$, $N = 2^n$.
- **Phase estimation (QPE):** n_c clock qubits in $|+\rangle$ control U^{2^j}, then the inverse QFT; an eigenphase $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$ gives the outcome $k \approx \varphi\, 2^{n_c}$.
- **HHL (as in `HHL.ipynb`):** registers clock (n_c), system (n_b), ancilla (1), total $n_c + n_b + 1$ qubits. $U = e^{iAt}$, so eigenvalue λ gives clock value $k = \lambda t\, 2^{n_c}/(2\pi)$. $C = 2\pi/(2^{n_c} t)$ is the smallest representable eigenvalue; for clock value k ($\tilde{\lambda}_k = kC$) the ancilla is rotated by $R_Y(\theta_k)$, $\theta_k = 2\arcsin(C/\tilde{\lambda}_k)$. Clock values $k \geq 2^{n_c-1}$ mean the negative value $k - 2^{n_c}$ (two's complement) unless a question says otherwise. The run succeeds when the ancilla is measured as 1 (and the clock as $0\ldots0$); $P_{\text{success}} = \sum_j |\beta_j|^2 C^2/\lambda_j^2$ when every λ_j is exactly representable.
- **Notebook default:** rows of A: $(2.5, -0.5, -1, 0)$, $(-0.5, 2.5, 0, -1)$, $(-1, 0, 2.5, -0.5)$, $(0, -1, -0.5, 2.5)$ (eigenvalues $1, 2, 3, 4$), $\vec{b} = (1, 0, 0, 0)$, $n_c = 4$, $t = \pi/8$, $C = 1$, 7 qubits, 20 000 shots. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).
- log means $\log_2$; $\tilde{O}$ hides polylogarithmic factors. Numerical answers are rounded.

SECTION A · Linear Algebra and Mathematical Foundations Q.1–20

Syllabus: matrices and vectors, linear systems and their classical solution, eigenvalues and eigenvectors, Hermitian and unitary matrices, spectral decomposition, sparse matrices, normalisation, matrix norms and the condition number.
Formats: Single correct (Q.1–19); Matching lists (Q.3–13); Assertion–Reason (Q.7–20); Statement I/II (Q.8–15); Paragraph based (Q.17–18). Each question: +3 / 0 / −1.

Q.1 The solution of the linear system $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is

(A) $(x_1, x_2) = (-0.2,\ 0.6)$ **(B)** $(x_1, x_2) = (0.6,\ 0.2)$
(C) $(x_1, x_2) = (1,\ 0.\overline{3})$ **(D)** $(x_1, x_2) = (0.2,\ 0.6)$

Q.2 The eigenvalues and the condition number κ of $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$ are

(A) 3 and 4; $\kappa = 4/3$ **(B)** 2 and 5; $\kappa = 2.5$
(C) 1 and 6; $\kappa = 6$ **(D)** 1 and 6; $\kappa = 7$

Q.3 Match each matrix (List-I) with the description that fits it (List-II).

List-I		List-II	
(P)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	**(1)**	Hermitian and unitary
(Q)	$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$	**(2)**	Hermitian, invertible, not unitary
(R)	$\begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix}$	**(3)**	Hermitian but singular
(S)	$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$	**(4)**	not diagonalisable

(A) P→2, Q→3, R→1, S→4 **(B)** P→1, Q→3, R→2, S→4
(C) P→1, Q→4, R→2, S→3 **(D)** P→1, Q→2, R→3, S→4

Q.4 The condition number of $A = \mathrm{diag}(1,\ 0.5,\ 0.01,\ 0.2)$ is

(A) 0.01 **(B)** 1.71 **(C)** 50 **(D)** 100

Q.5 The Hermitian matrix $A = \begin{pmatrix} 3 & 1-i \\ 1+i & 2 \end{pmatrix}$ has eigenvalues

(A) 3 and 2 **(B)** 0 and 5 **(C)** 1 and 4 **(D)** $2 \pm i$

Q.6 $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenpairs $\lambda_1 = 3$, $|u_1\rangle = \frac{1}{\sqrt{2}}(1,1)$ and $\lambda_2 = 1$, $|u_2\rangle = \frac{1}{\sqrt{2}}(1,-1)$. Using $A^{-1} = \sum_j \lambda_j^{-1} |u_j\rangle\langle u_j|$, the off-diagonal entry $(A^{-1})_{12}$ is

(A) $-\frac{1}{3}$ **(B)** $+\frac{1}{3}$ **(C)** $-\frac{2}{3}$ **(D)** $\frac{1}{6}$

Q.7 **Assertion (A):** Every Hermitian matrix can be diagonalised by a unitary matrix, $A = V\Lambda V^\dagger$.
Reason (R): All eigenvalues of a Hermitian matrix are real.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.8 **Statement I:** Every unitary matrix has condition number $\kappa = 1$.
Statement II: For every invertible matrix A, $\kappa(A^{-1}) = \kappa(A)$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.9 The 8×8 matrix of the 1-D discrete Laplacian has 2 on the diagonal, -1 on the first upper and lower off-diagonals and 0 elsewhere. Its sparsity s (maximum non-zeros per row) and its total number of non-zero entries are

(A) $s = 8$; 64 **(B)** $s = 3$; 22 **(C)** $s = 2$; 22 **(D)** $s = 3$; 24

Q.10 The eigenvalues of the $n \times n$ Laplacian of Q.9 (with 2 on the diagonal) are $\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$, $k = 1, \ldots, n$. For $n = 3$ the condition number is

(A) $3 + 2\sqrt{2} \approx 5.83$ **(B)** 2 **(C)** $\sqrt{2} \approx 1.41$ **(D)** $1 + \sqrt{2} \approx 2.41$

Q.11 The vector $\vec{b} = (3, 0, 4, 0)$ is amplitude-encoded in two qubits (index = binary number $q_0 q_1$). Measuring both qubits, the probability of the outcome $|10\rangle$ is

(A) 0.16 **(B)** 0.64 **(C)** 0.36 **(D)** 0.8

Q.12 With the eigenpairs of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ given in Q.6 and $\vec{b} = (1, 0)$, write $\vec{b} = \sum_j \beta_j |u_j\rangle$. Then $\vec{x} = \sum_j (\beta_j/\lambda_j) |u_j\rangle$ equals

(A) $(2,\ -1)$ **(B)** $\left(\frac{1}{2},\ -\frac{1}{2}\right)$ **(C)** $\left(\frac{2}{3},\ -\frac{1}{3}\right)$ **(D)** $\left(\frac{2}{3},\ \frac{1}{3}\right)$

Q.13 For $A = \mathrm{diag}(4,\ 1,\ 0.5,\ 3)$ match each quantity (List-I) with its value (List-II).

List-I		**List-II**	
(P)	spectral norm $\|A\|_2$	**(1)**	8
(Q)	spectral norm $\|A^{-1}\|_2$	**(2)**	4
(R)	condition number $\kappa(A)$	**(3)**	6
(S)	determinant $\det A$	**(4)**	2

(A) P→2, Q→4, R→3, S→1 **(B)** P→2, Q→3, R→1, S→4

(C) P→1, Q→4, R→2, S→3 **(D)** P→2, Q→4, R→1, S→3

Q.14 HHL needs a Hermitian matrix, so the non-Hermitian $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ is embedded as $\tilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}$ (4×4). The eigenvalues of $\tilde{A}$ are

(A) $3 \pm 2\sqrt{2}$ **(B)** ± 1, each twice

(C) $1, 1, 1, 1$ **(D)** $\pm(\sqrt{2}+1)$ and $\pm(\sqrt{2}-1)$

Q.15 **Statement I:** If A is Hermitian, then $U = e^{iAt}$ is unitary for every real t.
Statement II: If A is any real square matrix, then e^{iAt} is unitary for every real t.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.16 A linear system has $\kappa(A) = 1000$. The right-hand side is known with relative error $\|\delta\vec{b}\|/\|\vec{b}\| = 10^{-6}$. The best general upper bound on the relative error $\|\delta\vec{x}\|/\|\vec{x}\|$ of the solution is

(A) 10^{-3} **(B)** 10^{-6}

(C) 10^{-9} **(D)** $10^{-6}/\sqrt{1000} \approx 3 \times 10^{-8}$

Paragraph for Questions 17 and 18

The notebook `HHL.ipynb` lists the 2×2 example

$$A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The questions below concern the classical linear algebra that HHL relies on for this system.

Q.17 The eigenvalues of A and its condition number are

(A) $\frac{1}{3}$ and 1; $\kappa = 3$ **(B)** 1 and $-\frac{1}{3}$; $\kappa = 3$
(C) $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = \frac{4}{3}$ **(D)** $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$

Q.18 The solution $\vec{x} = A^{-1}\vec{b}$ and the probability of measuring the system qubit as $|0\rangle$ in the normalised state $|x\rangle$ are

(A) $\vec{x} = \left(1,\ \frac{1}{3}\right)$; $\frac{1}{\sqrt{10}}$ **(B)** $\vec{x} = \left(\frac{9}{8},\ \frac{3}{8}\right)$; 0.75
(C) $\vec{x} = \left(\frac{9}{8},\ \frac{3}{8}\right)$; 0.9 **(D)** $\vec{x} = \left(\frac{9}{8},\ -\frac{3}{8}\right)$; 0.9

Q.19 A 5×5 *non-Hermitian* matrix is prepared for HHL exactly as in the notebook: first the Hermitian embedding $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)$, then padding with an identity block to the next power of 2. The number of system qubits n_b is

(A) 5 **(B)** 4 **(C)** 10 **(D)** 3

Q.20 **Assertion (A):** For a Hermitian positive-definite matrix the condition number equals $\lambda_{\max}/\lambda_{\min}$.
Reason (R): The singular values of a Hermitian matrix are the absolute values of its eigenvalues.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

SECTION B · Quantum Computing Foundations Q.21–40

Syllabus: qubits and quantum states, superposition and the Born rule, entanglement, unitary and controlled gates, rotation gates, measurement and post-selection, the quantum Fourier transform and quantum phase estimation.
Formats: Single correct (Q.21–37); Matching lists (Q.24–36); Assertion–Reason (Q.31); Statement I/II (Q.30–38); Paragraph based (Q.39–40). Each question: +3 / 0 / −1.

Q.21 A qubit is in the state $|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2i\,|1\rangle)$. The probability of measuring 1 is

(A) $\frac{2}{\sqrt{5}}$ **(B)** $\frac{2}{5}$ **(C)** $\frac{4}{5}$ **(D)** $-\frac{4}{5}$

Q.22 The default HHL circuit uses 7 qubits. A state-vector simulator stores one complex double-precision number (16 bytes) per amplitude. The number of amplitudes and the memory of the state are

(A) 128 amplitudes; 2048 bytes **(B)** 7 amplitudes; 112 bytes
(C) 128 amplitudes; 1024 bytes **(D)** 49 amplitudes; 784 bytes

Q.23 The product HZH equals

(A) Y **(B)** Z **(C)** X **(D)** $-X$

Q.24 Match each rotation applied to $|0\rangle$ (List-I) with the resulting state (List-II).

List-I		List-II	
(P)	$R_Y(\pi)\,\|0\rangle$	**(1)**	$\|+\rangle$
(Q)	$R_Y(\pi/2)\,\|0\rangle$	**(2)**	$\frac{\sqrt{3}}{2}\|0\rangle + \frac{1}{2}\|1\rangle$
(R)	$R_Y(-\pi/2)\,\|0\rangle$	**(3)**	$\|1\rangle$
(S)	$R_Y(\pi/3)\,\|0\rangle$	**(4)**	$\|-\rangle$

(A) P→1, Q→3, R→4, S→2 **(B)** P→3, Q→1, R→4, S→2
(C) P→3, Q→4, R→1, S→2 **(D)** P→3, Q→1, R→2, S→4

Q.25 Which of the following two-qubit states is *entangled*?

(A) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ **(B)** $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
(C) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ **(D)** $\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$

Q.26 The control qubit is in $|+\rangle$ and the target in an eigenstate $|u\rangle$ of U with $U\,|u\rangle = e^{i\phi}\,|u\rangle$. After a controlled-$U$ the joint state is

(A) $|+\rangle \otimes e^{i\phi}\,|u\rangle$ (the phase stays on the target)
(B) $\frac{1}{\sqrt{2}}\Big(|0\rangle\,|u\rangle + |1\rangle\,U^\dagger\,|u\rangle\Big)$, entangled
(C) $\frac{1}{\sqrt{2}}\Big(e^{i\phi}\,|0\rangle + |1\rangle\Big) \otimes |u\rangle$
(D) $\frac{1}{\sqrt{2}}\Big(|0\rangle + e^{i\phi}\,|1\rangle\Big) \otimes |u\rangle$

Q.27 The inverse QFT on two qubits is applied to $|\psi\rangle = \frac{1}{2}\sum_{y=0}^{3} e^{2\pi i\, 3y/4}\,|y\rangle$. The result is

(A) $\frac{1}{2}\sum_y |y\rangle$ **(B)** $|10\rangle$ **(C)** $|01\rangle$ **(D)** $|11\rangle$

Q.28 Phase estimation with $n_c = 3$ clock qubits is run on an eigenstate with $\varphi = 0.375$. The measured clock bit string (c_0 = most significant bit) is

(A) 100 with probability 1
(B) 011 with probability 1
(C) 011 with probability ≈ 0.81
(D) 110 with probability 1

Q.29 Phase estimation with $n_c = 3$ clock qubits is run on an eigenstate with $\varphi = \frac{1}{3}$. The most likely outcome k and its probability are

(A) $k = 3;\ \approx 0.69$ **(B)** $k = 3;\ \approx 0.41$ **(C)** $k = 3;\ 1$ **(D)** $k = 2;\ \approx 0.69$

Q.30 **Statement I:** A unitary gate preserves the norm of every state vector.

Statement II: A projective measurement is reversible: the pre-measurement state can always be recovered from the post-measurement state.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.31 **Assertion (A):** The states $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ give identical statistics for every measurement.

Reason (R): The relative phase between the components of a superposition affects interference and hence measurement results.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.32 A controlled-$R_Y(\theta)$ with $\theta = 2\arcsin\frac{1}{2}$ acts on control $|1\rangle$ and target (ancilla) $|0\rangle$. The probability of then measuring the ancilla as 1 is

(A) $\frac{1}{4}$ **(B)** $\frac{3}{4}$ **(C)** $\frac{1}{16}$ **(D)** $\frac{1}{2}$

Q.33 A 3-qubit clock register $c_0c_1c_2$ (c_0 = most significant bit) controls a multi-controlled R_Y that fires only when all controls are 1. To make it act only on the clock value $k = 2$, the code places X gates before and after the rotation. Which X gates are needed?

(A) X on c_1 and c_2, before and after (4 gates)
(B) X on c_1 only, before and after (2 gates)
(C) X on c_0 and c_2, before and after (4 gates)
(D) none: $k = 2$ is selected by the rotation angle

Q.34 An ancilla a and a system register are in $\frac{1}{2}|0\rangle_a|\psi_0\rangle + \frac{\sqrt{3}}{2}|1\rangle_a|\psi_1\rangle$ ($|\psi_0\rangle, |\psi_1\rangle$ normalised). The ancilla is measured and the result 1 is kept (post-selection). The probability of this and the resulting system state are

(A) $\frac{3}{4}$; $|\psi_1\rangle$ **(B)** $\frac{\sqrt{3}}{2}$; $|\psi_1\rangle$
(C) $\frac{1}{4}$; $|\psi_0\rangle$ **(D)** $\frac{3}{4}$; $\frac{1}{\sqrt{2}}(|\psi_0\rangle + |\psi_1\rangle)$

Q.35 A circuit applies U_1, then U_2, then U_3 (all unitary). The circuit that undoes it applies

(A) $U_1^\dagger$, then $U_2^\dagger$, then $U_3^\dagger$
(B) U_3, then U_2, then U_1
(C) $U_1^\dagger U_2^\dagger U_3^\dagger$ applied as a single gate in the same order as the original
(D) $U_3^\dagger$, then $U_2^\dagger$, then $U_1^\dagger$

Q.36 The textbook QFT on $n = 4$ qubits (as used, inverted, on the clock register of the default HHL circuit). Match each quantity (List-I) with its value (List-II).

List-I		**List-II**	
(P)	number of Hadamard gates	**(1)**	2
(Q)	number of controlled-phase gates	**(2)**	$\pi/8$
(R)	number of SWAP gates	**(3)**	6
(S)	magnitude of the smallest controlled-phase angle	**(4)**	4

(A) P→3, Q→4, R→1, S→2 **(B)** P→4, Q→1, R→3, S→2
(C) P→4, Q→3, R→2, S→1 **(D)** P→4, Q→3, R→1, S→2

Q.37 A Hadamard is applied to each of four qubits in $|0000\rangle$. The amplitude of $|1011\rangle$ is

(A) $\frac{1}{16}$ **(B)** $\frac{1}{4}$ **(C)** $-\frac{1}{4}$ **(D)** $\frac{1}{\sqrt{8}}$

Q.38 **Statement I:** Every two-qubit pure state can be written as $|a\rangle \otimes |b\rangle$ for suitable single-qubit states $|a\rangle$, $|b\rangle$.

Statement II: For the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ the reduced state of each qubit is the maximally mixed state $\frac{1}{2}I$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 39 and 40

A unitary U has the eigenstate $|u\rangle$ with $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $\varphi = \frac{1}{4}$. Phase estimation uses $n_c = 2$ clock qubits c_0c_1 (c_0 = most significant bit). Both start in $|0\rangle$ and receive a Hadamard; c_1 (weight 2^0) controls U and c_0 (weight 2^1) controls U^2, so that the state becomes $\frac{1}{2}\sum_{k=0}^{3} e^{2\pi i\varphi k}|k\rangle|u\rangle$. Finally the inverse QFT is applied to the clock and the clock is measured.

Q.39 Before the inverse QFT, the amplitude of the clock state $|11\rangle$ (with $|u\rangle$ factored out) is

(A) $\frac{1}{2}$ **(B)** $-\frac{i}{2}$ **(C)** $-\frac{1}{2}$ **(D)** $\frac{i}{2}$

Q.40 After the inverse QFT the clock measurement gives

(A) 01 with probability 1, so $\tilde{\varphi} = \frac{1}{4}$

(B) 01 or 11 with probability $\frac{1}{2}$ each

(C) 11 with probability 1, so $\tilde{\varphi} = \frac{3}{4}$

(D) 10 with probability 1, so $\tilde{\varphi} = \frac{1}{2}$

SECTION C · HHL Algorithm Fundamentals Q.41–60

Syllabus: problem encoding, preparation of $|b\rangle$, Hamiltonian simulation e^{iAt}, phase estimation of the eigenvalues, eigenvalue inversion by a controlled rotation, the constant C, ancilla, uncomputation, measurement and the notebook's code.
Formats: Single correct (Q.41–57); Matching lists (Q.46–55); Assertion–Reason (Q.50); Statement I/II (Q.49–58); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 For a Hermitian A and a prepared $|b\rangle$, a successful run of the HHL algorithm produces

(A) the eigenvalues of A as a classical list

(B) the classical list of all d numbers $x_1, \ldots, x_d$

(C) the quantum state $|x\rangle \propto A^{-1}|b\rangle$ in the system register

(D) the matrix A^{-1} stored in the clock register

Q.42 With $|b\rangle = \sum_j \beta_j |u_j\rangle$ and exactly representable clock values k_j, the state after phase estimation (clock $\otimes$ system $\otimes$ ancilla) is

(A) $|0\ldots0\rangle\,|x\rangle\,|1\rangle$

(B) $\left(\frac{1}{\sqrt{d}}\sum_j |k_j\rangle\right) \otimes |b\rangle \otimes |0\rangle$

(C) $\sum_j \frac{\beta_j}{\lambda_j} |k_j\rangle\,|u_j\rangle\,|0\rangle$

(D) $\sum_j \beta_j |k_j\rangle\,|u_j\rangle\,|0\rangle$

Q.43 In the default notebook run ($n_c = 4$, $t = \pi/8$), the eigenvalue $\lambda = 3$ is written into the clock register as the bit string

(A) 0010 **(B)** 0110 **(C)** 0011 **(D)** 1100

Q.44 For the default notebook run ($n_c = 4$, $t = \pi/8$) the constant C and the rotation angle θ applied for the clock value of $\lambda = 2$ are

(A) $C = \frac{1}{16}$; $\theta = 2\arcsin\frac{1}{32}$

(B) $C = 1$; $\theta = \pi/3$

(C) $C = 1$; $\theta = \pi/6$

(D) $C = \pi/8$; $\theta = 2\arcsin(\pi/16)$

Q.45 With $C = 1$, the eigencomponent with $\lambda = 4$ leaves the ancilla in $\sqrt{1 - C^2/\lambda^2}\,|0\rangle + \frac{C}{\lambda}|1\rangle$. For this component the amplitude on $|1\rangle$ and the probability of measuring the ancilla as 1 are

(A) 4 and 1 **(B)** $\frac{1}{4}$ and $\frac{1}{4}$ **(C)** $\frac{1}{2}$ and $\frac{1}{4}$ **(D)** $\frac{1}{4}$ and $\frac{1}{16}$

Q.46 Match each part of the HHL circuit (List-I) with its role (List-II).

List-I		List-II	
(P)	clock register (n_c qubits)	**(1)**	is measured; outcome 1 flags a successful inversion
(Q)	system register (n_b qubits)	**(2)**	holds $\|b\rangle$ at the start and $\|x\rangle$ after success
(R)	ancilla qubit	**(3)**	holds the binary eigenvalue estimates k_j
(S)	inverse phase estimation after the rotation	**(4)**	returns the clock to $\|0\ldots0\rangle$ and removes its entanglement with the system

(A) P→3, Q→2, R→1, S→4

(B) P→2, Q→3, R→1, S→4

(C) P→3, Q→2, R→4, S→1

(D) P→3, Q→1, R→2, S→4

Q.47 HHL is run with $A = \mathrm{diag}(1, 2)$, $|b\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $C = 1$ and exactly representable eigenvalues. The success probability is

(A) 0.75 **(B)** 1 **(C)** 0.5 **(D)** 0.625

Q.48 An 8×8 Hermitian system is solved with the notebook's circuit and $n_c = 5$ clock qubits. The total number of qubits is

(A) 6 **(B)** 9 **(C)** 8 **(D)** 14

Q.49 **Statement I:** If the clock register is not uncomputed, the post-selected system register is in general not in the pure state $|x\rangle$, because it remains entangled with the clock.

Statement II: Uncomputation can be replaced by simply measuring the clock register, with the same output $|x\rangle$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.50 **Assertion (A):** In HHL the constant C must satisfy $C \leq \min_j |\lambda_j|$ (more precisely, $\leq$ the smallest $|\tilde{\lambda}|$ that can occur).

Reason (R): The controlled rotation puts the amplitude $C/\tilde{\lambda}$ on the ancilla state $|1\rangle$, and an amplitude cannot exceed 1 in magnitude.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.51 In the default notebook circuit ($n_c = 4$, $t = \pi/8$, $C = 1$, two's-complement clock), the clock value $k = 14$ (1110) is treated as the eigenvalue $\tilde{\lambda}$ and rotated by θ, where

(A) $\tilde{\lambda} = -2,\ \theta = +\pi/3$ **(B)** $\tilde{\lambda} = 2,\ \theta = \pi/3$

(C) $\tilde{\lambda} = -2,\ \theta = -\pi/3$ **(D)** $\tilde{\lambda} = 14,\ \theta = 2\arcsin\frac{1}{14}$

Q.52 The eigenvalue-inversion loop of `HHL.ipynb` runs over $k = 1, \ldots, 2^{n_c} - 1$; for each k it applies `kernel.x` to every clock qubit whose bit of k is 0, then the multi-controlled `cry`, then the same `x` gates again. For $n_c = 4$ the total number of `x` gates is

(A) 56 **(B)** 28 **(C)** 60 **(D)** 32

Q.53 Let $\vec{a}$ be the (unnormalised) amplitudes of the system register in the branch clock $= 0 \ldots 0$, ancilla $= 1$ of an exact HHL run. The notebook recovers the solution, including its norm, as

(A) $\vec{x} = \dfrac{\vec{a}}{\sqrt{P_{\text{success}}}}$ **(B)** $\vec{x} = \dfrac{\|\vec{b}\|}{C}\,\vec{a}$

(C) $\vec{x} = \dfrac{C}{\|\vec{b}\|}\,\vec{a}$ **(D)** $\vec{x} = \vec{a}$

Q.54 For the notebook's default system (A with eigenvalues $1, 2, 3, 4$ and $\vec{b} = (1, 0, 0, 0)$, see conventions) the exact solution is

(A) $\vec{x} = \frac{1}{48}(25,\ 7,\ 11,\ 5)$
(B) $\vec{x} = \frac{1}{48}(25,\ -7,\ -11,\ 5)$
(C) $\vec{x} = \frac{1}{\sqrt{820}}(25,\ 7,\ 11,\ 5)$
(D) $\vec{x} = (1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4})$

Q.55 Match each step of HHL (List-I) with the CUDA-Q / Python code from `HHL.ipynb` that implements it (List-II).

List-I		**List-II**	
(P)	amplitude encoding of $\|b\rangle$	**(1)**	`kernel.cry(theta, clock, ancilla)`
(Q)	controlled $U^{2^j} = e^{iAt2^j}$	**(2)**	`cudaq.register_operation(name, matrix_power(U, 2**j))`
(R)	eigenvalue inversion	**(3)**	`bits[:n_c] == '0'*n_c and bits[-1] == '1'`
(S)	post-selection of the shots	**(4)**	`kernel.qalloc(initial_state_vector)`

(A) P→4, Q→1, R→2, S→3
(B) P→4, Q→2, R→3, S→1
(C) P→2, Q→4, R→1, S→3
(D) P→4, Q→2, R→1, S→3

Q.56 Starting from $|00\rangle$, which circuit prepares $|b\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$, i.e. $\vec{b} \propto (1, -1, 1, -1)$?

(A) H on both qubits only
(B) H on both qubits, then Z on q_1
(C) H on both qubits, then Z on q_0
(D) X on both qubits, then H on both qubits

Q.57 Inside the phase-estimation part of HHL, the inverse QFT on the clock register

(A) computes $1/\lambda$ for every eigenvalue
(B) turns the kicked-back phases $e^{2\pi i\, kj/2^{n_c}}$ into the computational-basis state $|k\rangle$ (the binary eigenvalue estimate)
(C) measures the eigenvalues and sends them to the classical computer
(D) prepares $|b\rangle$ in the eigenbasis of A

Q.58 Assume $|b\rangle$ is itself an eigenvector $|u_j\rangle$ of A and λ_j is exactly representable by the clock.

Statement I: The success probability is C^2/λ_j^2.

Statement II: After phase estimation the clock register is in the single basis state $|k_j\rangle$ (not entangled with the system).

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 59 and 60

HHL is run by hand on $A = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$, $\vec{b} = (1, 0)$ ($n_b = 1$). The eigenpairs are $\lambda_1 = 1$, $|u_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and $\lambda_2 = 2$, $|u_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Phase estimation uses $n_c = 2$ clock qubits and $t = \pi/2$; because both eigenvalues are positive, the clock value k is read as an *unsigned* integer here ($\tilde{\lambda}_k = kC$). States are written as $|\text{clock}\rangle\,|\text{system}\rangle$; the ancilla starts in $|0\rangle$ and is not shown until the rotation.

Q.59 The state after phase estimation is

(A) $\frac{1}{\sqrt{2}}\left(|00\rangle\,|u_1\rangle + |01\rangle\,|u_2\rangle\right)$

(B) $\frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \otimes |0\rangle$

(C) $\frac{1}{\sqrt{2}}\left(|10\rangle\,|u_1\rangle + |01\rangle\,|u_2\rangle\right)$

(D) $\frac{1}{\sqrt{2}}\left(|01\rangle\,|u_1\rangle + |10\rangle\,|u_2\rangle\right)$

Q.60 After the controlled rotation, uncomputation and a successful ancilla measurement, the success probability and the system state are

(A) $P = \frac{5}{8}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle + |1\rangle)$

(B) $P = \frac{3}{4}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$

(C) $P = \frac{5}{8}$; $\frac{1}{\sqrt{10}}(3\,|0\rangle - |1\rangle)$

(D) $P = 1$; $|0\rangle$

SECTION D · Advanced HHL and Implementation Q.61–80

Syllabus: complexity, sparsity and condition-number dependence, error analysis, post-selection and amplitude amplification, circuit resources, CUDA-Q implementation and memory, noise, benchmarking of recorded results, limitations, dequantisation and applications.
Formats: Single correct (Q.61–78); Matching lists (Q.63–74); Assertion–Reason (Q.65–77); Statement I/II (Q.70); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 For an s-sparse Hermitian $N \times N$ matrix with condition number κ, target error ϵ, and efficient preparation of $|b\rangle$, the run time of the original HHL algorithm (2009) scales as

(A) $O(\log N)$, independent of κ and ϵ
(B) $\tilde{O}(N\, s\, \kappa)$
(C) $\tilde{O}(\log(N)\, s^2 \kappa^2/\epsilon)$
(D) $O(N^3)$

Q.62 Suppose $P_{\text{success}} = 1/\kappa^2$ with $\kappa = 10$. The expected number of repetitions needed for one successful run without, and with, amplitude amplification is about

(A) 1 without; 1 with
(B) 100 without; 10 with
(C) 100 without; 100 with (no improvement is possible)
(D) 10 without; $\sqrt{10}$ with

Q.63 Match each quantum linear-system algorithm (List-I) with its characteristic improvement or complexity (List-II).

List-I		List-II	
(P)	HHL (Harrow, Hassidim, Lloyd, 2009)	**(1)**	precision dependence polylog$(1/\epsilon)$ via Fourier/Chebyshev fitting of $1/x$ (LCU)
(Q)	Variable-time amplitude amplification (Ambainis, 2012)	**(2)**	almost linear dependence on κ
(R)	Childs, Kothari and Somma (2017)	**(3)**	$\tilde{O}(\kappa^2/\epsilon)$ dependence on κ and ϵ
(S)	Discrete adiabatic method (Costa et al., 2022)	**(4)**	optimal $O(\kappa \log(1/\epsilon))$ query complexity

(A) P→2, Q→3, R→1, S→4
(B) P→3, Q→2, R→4, S→1
(C) P→3, Q→2, R→1, S→4
(D) P→3, Q→1, R→2, S→4

Q.64 Why can HHL *not* efficiently output all d components of $\vec{x}$ for $d = 2^{20}$?

(A) The no-cloning theorem forbids measuring the system register
(B) Sampling gives each $|x_i|^2$ only statistically; estimating all d values needs $\Omega(d)$ repetitions (about d/ϵ^2 for precision ϵ), destroying the $\log d$ scaling
(C) Complex amplitudes cannot be measured at all
(D) The ancilla measurement destroys $|x\rangle$ every time, so the system register never contains $|x\rangle$

Q.65 **Assertion (A):** HHL gives an exponential speed-up for solving any linear system $A\vec{x} = \vec{b}$ and printing the solution $\vec{x}$.

Reason (R): Under suitable assumptions (sparse, well-conditioned A, efficient preparation of $|b\rangle$, only a property of $|x\rangle$ wanted) the run time of HHL is polylogarithmic in the dimension.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.66 The notebook is run with `NUM_CLOCK_QUBITS = 3` on the default matrix (eigenvalues $1, 2, 3, 4$). Since $\kappa = 4$ exceeds the $2^{n_c-1} - 1 = 3$ positive levels, "auto" sets $t = 2\pi \cdot 3/(2^{n_c}\lambda_{\max})$. The ideal clock values $k_j = \lambda_j t 2^{n_c}/2\pi$ are

(A) 0.75, 1.5, 2.25, 3: only $\lambda = 4$ is exactly representable

(B) 1, 2, 3, 4: all exact

(C) 0.5, 1, 1.5, 2: $\lambda = 2$ and 4 exact

(D) 1.5, 3, 4.5, 6: aliasing of $\lambda = 3, 4$

Q.67 Using the rule of thumb $n_c \approx \lceil \log_2(\kappa/\epsilon) \rceil$ for the number of clock qubits, a system with $\kappa = 100$ and target relative precision $\epsilon = 0.01$ needs about

(A) 7 clock qubits

(B) 10 clock qubits

(C) 100 clock qubits

(D) 14 clock qubits

Q.68 The notebook's state-vector approach is used for a Hermitian $2^{20} \times 2^{20}$ matrix with $n_c = 10$ (fp64, 16 bytes per amplitude). The number of qubits and the memory of the state vector are

(A) 31 qubits; ≈ 17.2 GB

(B) 31 qubits; 496 bytes

(C) $2^{20} + 11$ qubits; beyond any computer

(D) 31 qubits; ≈ 34.4 GB

Q.69 e^{iAt} with $A = A_1 + A_2$ ($[A_1, A_2] \neq 0$) is approximated by r first-order Trotter steps $\left(e^{iA_1 t/r} e^{iA_2 t/r}\right)^r$. To keep the total error below ϵ, r must grow as

(A) $r = O(t^2 \|[A_1, A_2]\|/\epsilon)$

(B) $r = O(\log(1/\epsilon))$, independent of t

(C) $r = O(t/\epsilon^2)$

(D) $r = 1$ always suffices, because the error cancels

Q.70 About `HHL.ipynb`:

Statement I: The controlled evolutions e^{iAt2^j} are implemented as exact matrix exponentials (`scipy.linalg.expm`) registered as custom gates, not by Hamiltonian simulation.

Statement II: The notebook executes HHL on quantum-hardware targets when `BACKEND = "hardware"`.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.71 In the recorded run the ideal success probability is 0.355903 and 7086 of 20 000 shots (0.3543) were accepted. The binomial standard error of the measured fraction and the verdict are

(A) ≈ 0.0007; the deviation is $\approx 2.3\sigma$

(B) ≈ 0.0034; the deviation is $\approx 4.7\sigma$, a circuit error

(C) ≈ 0.0034; the deviation is $\approx 0.5\sigma$, consistent with shot noise

(D) ≈ 0.012; the deviation is $\approx 0.1\sigma$

Q.72 In the recorded run, the post-selected shots (7086 accepted) estimate $|x_0|^2/\|\vec{x}\|^2$ as 0.7691; the exact value is 0.7622. The binomial standard error of this estimate and the deviation in units of it are

(A) ≈ 0.0051; $\approx 1.4\sigma$

(B) ≈ 0.012; $\approx 0.6\sigma$, the exact standard error

(C) ≈ 0.0051; $\approx 14\sigma$

(D) ≈ 0.0030; $\approx 2.3\sigma$

Q.73 The default circuit ($n_c = 4$, 4×4 matrix) has 15 multi-controlled rotations and 8 controlled $e^{\pm iAt2^j}$ gates on 7 qubits. With the same matrix and $n_c = 5$ the notebook's circuit has

(A) 31 rotations, 10 controlled $e^{\pm iAt2^j}$, 7 qubits

(B) 31 rotations, 10 controlled $e^{\pm iAt2^j}$, 8 qubits

(C) 16 rotations, 10 controlled $e^{\pm iAt2^j}$, 8 qubits

(D) 31 rotations, 5 controlled $e^{\pm iAt2^j}$, 8 qubits

Q.74 Match each error source (List-I) with its main effect on HHL (List-II).

List-I		**List-II**	
(P)	bit flip on the ancilla just before measurement	**(1)**	statistical error $\sim 1/\sqrt{N_{\text{acc}}}$ in the estimated $\lvert x_i\rvert^2$
(Q)	phase error on a clock qubit during phase estimation	**(2)**	wrong eigenvalue estimate, so a wrong factor $1/\tilde{\lambda}$ is applied
(R)	imperfect uncomputation of the clock	**(3)**	failed runs are accepted (and good ones rejected): biased P_{success} and contaminated output
(S)	finite number of shots	**(4)**	residual clock–system entanglement: the output is a mixed state

(A) P→3, Q→2, R→1, S→4

(B) P→2, Q→3, R→4, S→1

(C) P→3, Q→4, R→2, S→1

(D) P→3, Q→2, R→4, S→1

Q.75 Which output can HHL provide efficiently (under its assumptions)?

(A) The full inverse matrix A^{-1}

(B) An estimate of an expectation value $\langle x|M|x\rangle$ for an efficiently measurable observable M

(C) The determinant of A

(D) The complete vector $\vec{x}$ to 16 significant digits

Q.76 "Dequantisation" results (Tang 2019 and successors) give classical algorithms with run time polylogarithmic in d for linear systems when

(A) A is Hermitian

(B) A has low rank and ℓ^2-norm sample-and-query access to A and $\vec{b}$ is available

(C) $\vec{b}$ is a computational basis vector

(D) A is sparse and well conditioned with $\kappa = \text{polylog}(d)$, and $\vec{b}$ is given as an efficient circuit

Q.77 **Assertion (A):** At fixed N, s and ϵ, doubling the condition number roughly quadruples the run time of the original HHL algorithm.

Reason (R): The original HHL run time scales as $\tilde{O}(\log(N)s^2\kappa^2/\epsilon)$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.78 For a non-Hermitian invertible A the notebook solves $\begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}\begin{pmatrix} \vec{y} \\ \vec{z} \end{pmatrix} = \begin{pmatrix} \vec{b} \\ \vec{0} \end{pmatrix}$. The solution satisfies

(A) $\vec{y} = A^{-1}\vec{b}$ (first half) and $\vec{z} = \vec{0}$

(B) $\vec{y} = \vec{z} = A^{-1}\vec{b}/\sqrt{2}$

(C) $\vec{y} = A^{-1}\vec{b}$ and $\vec{z} = (A^\dagger)^{-1}\vec{b}$

(D) $\vec{z} = A^{-1}\vec{b}$ (second half) and $\vec{y} = \vec{0}$

Paragraph for Questions 79 and 80

The previously recorded default run of `HHL.ipynb` (7 qubits, `qpp-cpu`) printed:

```
Success probability:  state 0.355903, shots 0.3543 (7086 of 20000), ideal 0.355903
Fidelity |<x_classical|x_quantum>|^2 :  1.0000000000
Relative error of the recovered vector:  1.655e-15
```

In the eigenbasis of the default A (eigenvalues $1, 2, 3, 4$) the vector $|b\rangle = |00\rangle$ has all four coefficients $\beta_j = \frac{1}{2}$; $C = 1$ and $\|\vec{b}\| = 1$.

Q.79 The exact value of the ideal success probability 0.355903 is

(A) $\dfrac{205}{576}$ **(B)** $\dfrac{25}{48}$ **(C)** $\dfrac{205}{144}$ **(D)** $\dfrac{1}{4}$

Q.80 Without knowing $\vec{x}$ in advance, the norm $\|\vec{x}\|$ can be inferred from the success probability. Its value is

(A) $\dfrac{24}{\sqrt{205}} \approx 1.676$

(B) $\dfrac{205}{576} \approx 0.356$

(C) $\dfrac{\sqrt{205}}{24} \approx 0.597$

(D) 1, because quantum states are normalised

RESPONSE SHEET

Darken exactly ONE bubble per question. Leave all four bubbles empty to skip a question (0 marks).

Name: _____ **Roll No.:** _____

Section A	Section B	Section C	Section D
1 (A)(B)(C)(D)	21 (A)(B)(C)(D)	41 (A)(B)(C)(D)	61 (A)(B)(C)(D)
2 (A)(B)(C)(D)	22 (A)(B)(C)(D)	42 (A)(B)(C)(D)	62 (A)(B)(C)(D)
3 (A)(B)(C)(D)	23 (A)(B)(C)(D)	43 (A)(B)(C)(D)	63 (A)(B)(C)(D)
4 (A)(B)(C)(D)	24 (A)(B)(C)(D)	44 (A)(B)(C)(D)	64 (A)(B)(C)(D)
5 (A)(B)(C)(D)	25 (A)(B)(C)(D)	45 (A)(B)(C)(D)	65 (A)(B)(C)(D)
6 (A)(B)(C)(D)	26 (A)(B)(C)(D)	46 (A)(B)(C)(D)	66 (A)(B)(C)(D)
7 (A)(B)(C)(D)	27 (A)(B)(C)(D)	47 (A)(B)(C)(D)	67 (A)(B)(C)(D)
8 (A)(B)(C)(D)	28 (A)(B)(C)(D)	48 (A)(B)(C)(D)	68 (A)(B)(C)(D)
9 (A)(B)(C)(D)	29 (A)(B)(C)(D)	49 (A)(B)(C)(D)	69 (A)(B)(C)(D)
10 (A)(B)(C)(D)	30 (A)(B)(C)(D)	50 (A)(B)(C)(D)	70 (A)(B)(C)(D)
11 (A)(B)(C)(D)	31 (A)(B)(C)(D)	51 (A)(B)(C)(D)	71 (A)(B)(C)(D)
12 (A)(B)(C)(D)	32 (A)(B)(C)(D)	52 (A)(B)(C)(D)	72 (A)(B)(C)(D)
13 (A)(B)(C)(D)	33 (A)(B)(C)(D)	53 (A)(B)(C)(D)	73 (A)(B)(C)(D)
14 (A)(B)(C)(D)	34 (A)(B)(C)(D)	54 (A)(B)(C)(D)	74 (A)(B)(C)(D)
15 (A)(B)(C)(D)	35 (A)(B)(C)(D)	55 (A)(B)(C)(D)	75 (A)(B)(C)(D)
16 (A)(B)(C)(D)	36 (A)(B)(C)(D)	56 (A)(B)(C)(D)	76 (A)(B)(C)(D)
17 (A)(B)(C)(D)	37 (A)(B)(C)(D)	57 (A)(B)(C)(D)	77 (A)(B)(C)(D)
18 (A)(B)(C)(D)	38 (A)(B)(C)(D)	58 (A)(B)(C)(D)	78 (A)(B)(C)(D)
19 (A)(B)(C)(D)	39 (A)(B)(C)(D)	59 (A)(B)(C)(D)	79 (A)(B)(C)(D)
20 (A)(B)(C)(D)	40 (A)(B)(C)(D)	60 (A)(B)(C)(D)	80 (A)(B)(C)(D)

Score computation (filled in by the evaluator, using the Answer Book)

Sec.	**Topic**	**Attempted**	**Correct N_c**	**Wrong N_w**	**Marks $3N_c - N_w$**
A	Linear Algebra and Mathematical Foundations				
B	Quantum Computing Foundations				
C	HHL Algorithm Fundamentals				
D	Advanced HHL and Implementation				
				Total (out of 240)	